

4(a)	<i>diabetes:</i> insulin from (GM) bacteria ; A ref to, yeast / animal cells <i>single gene disorder:</i> gene therapy or insert, functional / normal / dominant, allele into, target / somatic / stem, cells or genome ; A gene R <i>replacing one allele with another</i>	2
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Question	Answer	Marks																
4(b)(i)	any four from: <i>valid</i> 1 GM spinach decreases number of, bacteria / <i>E.coli</i> or GM spinach has lowest number of, bacteria / <i>E.coli</i> ; 2 at all times (meat is stored) or the longer the time stored the more effective it is or most effective after 72 hours ; 3 data quote comparing no treatment or buffer alone with GM spinach extract at stated time ; <table><tr><td>time / hrs</td><td>no treatment</td><td>buffer alone</td><td>buffer plus GM spinach</td></tr><tr><td>1</td><td>3.7–3.8</td><td>3.8–3.9</td><td>1.4–1.5</td></tr><tr><td>24</td><td>3.2–3.3</td><td>3.5–3.6</td><td>0.5–0.6</td></tr><tr><td>72</td><td>2.6–2.7</td><td>3.3–3.4</td><td>0.1–0.2</td></tr></table> <i>not valid</i> 4 no treatment / buffer alone, (also) decrease number of, bacteria / <i>E.coli</i> or GM spinach does not kill all, bacteria / <i>E.coli</i> ; 5 meat should not normally be left at 10°C or people unlikely to eat meat left at, 10°C / for 72 hours ; 6 does not test effect on people / AW ;	time / hrs	no treatment	buffer alone	buffer plus GM spinach	1	3.7–3.8	3.8–3.9	1.4–1.5	24	3.2–3.3	3.5–3.6	0.5–0.6	72	2.6–2.7	3.3–3.4	0.1–0.2	4
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Question	Answer	Marks
4(b)(ii)	any three from: 1 may alter, taste / texture / quality, of meat ; 2 may be unsafe for humans / allergies / side effects ; 3 may harm, gut / beneficial, bacteria or may increase growth of, non-colicin sensitive / other, bacteria ; 4 contaminate food labelled as organic / AW ; 5 <i>E.coli</i> / bacteria, may develop resistance to, (spray / colicins) ; R GM spray / colicins, cause mutation 6 AVP ; e.g. may decrease food hygiene practices in, meat factories / abattoirs / restaurants / food outlets ;	3

Question	Answer	Marks
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